3.1 → Risk classification

3.1.1 → Importance of equitable rates

→ Goal is to segment in order to charge the true rates to each group in the population, thus avoiding adverse selection.

→ Adverse selection occurs when you charge the same rate instead of segmenting

⇒ As a result, over/under charge certain groups + over time start attracting more & more riskier policies

⇒ unprofitable
⇒ rate changes
(... repeat ...) → until either:
1) segment in a more refined manner,
2) become insolvent, or
3) specialize in insuring high-risk insureds only

→ opposite of this is favorable selection.

→ "skimming the cream" is using risk characteristics to identify, attract, + select the lower-risk insureds
(marketing) (underwriting)

3.1.2 → Criteria for evaluating rating variables

→ The first step in classification ratemaking is to identify the rating variables that will be used to classify risks. The criteria for evaluating the appropriateness of rating variables can be grouped into four categories:

→ 1) Statistical criteria
→ a) Statistical significance → different levels of the rating variable should have different expected costs that are statistically significant & stable over time
→ b) Homogeneity → the risks within the same level should have similar expected costs
→ c) Credibility → Each level should be large enough to allow credible estimate of costs
balancing act in terms of how fine to make the groups

→ reasons to not include a variable are just the negative or opposite of these

→ 2) Operational criteria
→ a) Objective → clear + measurable
→ b) Inexpensive to administrate → cost to obtain shouldn't outweigh potential benefit
→ c) Verifiable → not easily manipulated + easily verifiable

→ 3) Social criteria
→ a) Affordability → Insurance should be affordable for all risks (even high-risks) ↳ can offer less coverage
→ b) Causality → A clear cause-&-effect between a rating variable + expected losses will increase the social acceptability of the classification
→ c) Controllability → policy holders generally prefer to have some degree of control over the class to which they belong & the ability to influence the premium charged (e.g. in home insurance, whether or not a security system is installed)
→ d) Privacy concerns → not infringe on privacy of insured
balancing act in practice

→ 4) Legal criteria → follows applicable laws

→ Classification ratings → groups risks w/ similar characteristics / loss propensity → appropriate when rating a homogeneous group of risks } → Both use historical data + risk characteristics to determine a rate

→ Individual risk rating → Tailored to the individual insured

→ Standard rates set by the state use neither rating plan

→ wording of question ⇒ "EVALUATE" means DESCRIBE + RECOMMEND (yay/nay)

3.1.3 → Univariate classification

→ Goal → Determine relativities for each level of a rating variable ⇒ i.e) determines rate differentials by analyzing historical experience of each level of a rating variable independently
→ There are three approaches

→ Pure premium approach
→ The basic pure premium approach determines the indicated relativities by comparing the expected pure premiums for each of the levels w/in a rating variable.
→ when using the pure premium approach, the need to trend & develop the loss & ALAE depends on the nature of the portfolio.
→ Data should be adjusted for extraordinary + catastrophic events prior to a classification analysis.
→ The table below shows the indicated relativity for each territory using the pure premium approach

(1) Territory	(2) Exposures	(3) Loss & ALAE	(4) Indicated Pure Premium	(5) Indicated Relativity	(6) Indicated Relativity to Base
1	300	$15,698.08	$52.33	0.7696	**0.7231**
2	390	$28,221.07	$72.36	1.0643	**1.0000**
3	310	$24,072.96	$77.65	1.1421	**1.0731**
Total	1,000	$67,992.11	$67.99	1.0000	0.9396

(5) = (4i)/(4 total); (6) = (5i)/(5₂) or (4i)/(4₂)

→ Loss ratio approach
→ The loss ratio approach determines the indicated relativities by comparing the LRs for each of the levels w/in a rating variable. When using this approach, EP should be brought to the current level for each class of risks.
→ The table below shows the indicated relativity for each territory using the LR approach.

(1) Territory	(2) Premium at Current Rate Level	(3) Loss & ALAE	(4) Loss & ALAE Ratio	(5) Indicated Change Factor	(6) Current Relativity	(7) Indicated Relativity	(8) Indicated Relativity to Base
1	$25,740.66	$15,698.08	60.99%	1.0763	0.6000	0.6458	**0.6354**
2	$49,002.80	$28,221.07	57.59%	1.0164	1.0000	1.0164	**1.0000**
3	$45,256.64	$24,072.96	53.19%	0.9388	1.3000	1.2204	**1.2007**
Total	$120,000.10	$67,992.11	56.66%	1.0000			

(5) = (4i)/(4 total) → factor to adjust now; (7) = (5)×(6); (8) = (7i)/(7₂)

→ Adjusted pure premium approach

→ The LR approach requires premium at CRL for each level of the rating variable, which may not always be available or practical to obtain. In such cases, it is necessary to use the pure premium approach. To mitigate the impact of any distributional bias, the pure premium approach can be performed using exposures adjusted by the exposure-weighted average relativity of all other variables.

→ The exposure-weighted AOI relativity for each territory is calculated as follows:

AOI	Charged AOI Factor	Exposures by Territory 1	2	3
Low	0.80	10	130	150
Medium	1.00	110	120	120
High	1.36	180	140	40
Wtd Avg AOI Relativity by Territory		1.2093	1.0626	0.9497

→ For example, the weighted average AOI relativity of territory 1 is

$$\frac{10(0.80) + 110(1.00) + 180(1.36)}{10 + 110 + 180} = 1.2093$$

→ The table below shows the indicated relativity for each territory using the PP approach w/ adjusted exposures

(1) Territory	(2) Exposures	(3) Wtd Avg AOI Relativity	(4) Adjusted Exposures	(5) Loss & ALAE	(6) Indicated Adjusted Pure Premium	(7) Indicated Relativity	(8) Indicated Relativity to Base
1	300	1.2093	362.8	$15,698.08	$43.27	0.6820	**0.6354**
2	390	1.0626	414.4	$28,221.07	$68.10	1.0733	**1.0000**
3	310	0.9497	294.4	$24,072.96	$81.77	1.2887	**1.2007**
Total	1,000		1,071.6	$67,992.11	$63.45	1.0000	0.9317

(4) = (2)×(3); (6) = (5)/(4); (7) = (6i)/(6 total); (8) = (7i)/(7₂)

→ As the exposures have been adjusted using the current amount of insurance relativities charged, the resulting indicated relativities are the same as those calculated using the LR approach. As w/ the LR approach, the remaining distortions in the indicated relativities come from the variation between the true AOI relativities + the AOI relativities charged by the insurer.

⇒ LR approach results = Adj PP approach results
↳ need premium @ CRL for each level

→ Potential distortions/shortcomings → Univariate methods have an inability to accurately capture the effect of other rating variables
→ Thus, many companies use alternative methods (like multivariate classification)

Adjusted pure premium approach

(0) factor	(1) Adj exposures	(2) loss + LAE	(3) Indicated adj PP	(4) Indicated relativity	(5) Indicated relativity to base
factor of interest	sumproduct of current factor over exp by other factor		(2)/(1)	(3)/(3 total)	(4)/(4 base)

→ Changing class relativities

→ Ind class rate change = Ind avg rate change factor × Rel change factor × OBF − 1
Ind avg rate change factor ↓ =1 if revenue neutral
Rel change factor ↓ cred-wtd + rebased
OBF → accounts for the change in the average relativity that we unintentionally cause by creating new class relativities for each class. We need to avoid changing the average relativity since that would also change the average rate. Our goal is simply to "rework" the relativities so that the amount that each class pays is proportional to their expected cost.

3.1.4 → Univariate classification w/ credibility

→ GOAL → Using credibility enhances the reliability of estimates derived from univariate analysis
→ Blending class-specific data w/ broader, more stable datasets mitigates the impact of random variations + ensures more accurate rate indications

→ Credibility estimate = Credibility × Experience + (1 − Credibility) × complement
need same base

→ Pure premium + adjusted pure premium approaches →

(note for these examples, the complement of credibility is no change)

calculated w/ a given formula

(1) Territory	(2) Exposures	(3) Current Relativity	(4) Normalized Current Relativity	(5) Indicated Relativity to Total	(6) Credibility	(7) Cred-Wtd Indicated Relativity	(8) Cred-Wtd Indicated Relativity to Base
1	300	0.600	0.6166	0.7696	0.7746	0.7351	**0.6935**
2	390	1.000	1.0277	1.0643	0.8832	1.0600	**1.0000**
3	310	1.300	1.3361	1.1421	0.7874	1.1833	**1.1164**
Total	1,000	0.9730	1.0000	1.0000			

- (3)total = [(3) × (2)] ÷ (2)total
- (4) = (3) ÷ (3)total
- (7) = (6) × (5) + [1 - (6)] × (4)
- (8) = (7) ÷ (7)base

If adjusted PP method, use adjusted exposures instead ⇒ rest is same

→ to rebase a set of relativities to the total:
→ 1) Calculate the weighted avg of the relativities
→ 2) Divide each relativity by the avg calculated in (1)
→ 3) The re-based relativities should have a weighted-avg = 1

→ Loss ratio approach

(1) Territory	(2) Current Relativity	(3) Indicated Relativity	(4) Credibility	(5) Cred-Wtd Indicated Relativity	(6) Cred-Wtd Indicated Relativity to Base
1	0.600	0.6458	0.7746	0.6355	**0.6264**
2	1.000	1.0164	0.8832	1.0145	**1.0000**
3	1.300	1.2204	0.7874	1.2373	**1.2197**

- (5) = (4) × (3) + [1 - (4)] × (2)
- (6) = (5) ÷ (5)base

OR

(instead of ind relativity)

(1) Territory	(2) Current Relativity	(3) Indicated Change Factor	(4) Credibility	(5) Cred-Wtd Indicated Change Factor	(6) Cred-Wtd Indicated Relativity	(7) Cred-Wtd Indicated Relativity to Base
1	0.600	1.0763	0.7746	1.0591	0.6355	**0.6264**
2	1.000	1.0164	0.8832	1.0145	1.0145	**1.0000**
3	1.300	0.9388	0.7874	0.9518	1.2373	**1.2197**

- (5) = (4) × (3) + [1 - (4)] × 1.0000
- (6) = (5) × (2)
- (7) = (6) ÷ (6)base

Alternative solution ~ shorter

→ If we were to rebase the relativities in the LR approach, here is what it would look like:

(1) Territory	(2) Premium at Current Rate Level	(3) Premium at CRL at Base	(4) Current Relativity	(5) Normalized Current Relativity	(6) Indicated Relativity	(7) Normalized Indicated Relativity	(8) Credibility	(9) Cred-Wtd Indicated Relativity	(10) Cred-Wtd Indicated Relativity to Base
1	$25,740.66	$42,901.10	0.600	0.6336	0.6458	0.6820	0.7746	0.6710	**0.6264**
2	$49,002.80	$49,002.80	1.000	1.0560	1.0164	1.0733	0.8832	1.0713	**1.0000**
3	$45,256.64	$34,812.80	1.300	1.3728	1.2204	1.2887	0.7874	1.3066	**1.2197**
Total	$120,000.10	$126,716.70	0.9470	1.0000	0.9470	1.0000			

- (3) = (2) ÷ (4)
- (4)total = [(4) × (3)] ÷ (3)total
- (5) = (4) ÷ (4)total
- (6)total = [(6) × (3)] ÷ (3)total
- (7) = (6) ÷ (6)total
- (9) = (8) × (7) + [1 - (8)] × (5)
- (10) = (9) ÷ (9)base

→ credibility is not based on adj exposures when calculating relativities
→ have to normalize current factors before using them in a credibility calculation
↳ wtd avg rel = sumproduct (current factor, adj exp) / sum (adj exp)
⇒ normalized rel = current rel / wtd avg rel

3.2 → Multivariate classification

3.2.1 → Minimum bias procedures

→ Minimum bias procedures are iteratively standardized univariate approaches that account for an uneven mix of business. Each procedure involves the following steps:
→ 1) Select a rating structure, such as additive, multiplicative, or combined
→ 2) Choose a bias function, which measures the difference between the procedure's observed loss statistics & indicated loss statistics. Examples of bias functions are balance principle, least squares, χ^2, + maximum likelihood functions.
→ 3) Weighting both sides of the bias function by the exposures in each cell to adjust for uneven mix of business.
→ 4) "Minimizing the bias", which is accomplished by equating the sum of indicated pure premium + the sum of the weighted observed loss costs for every level of every rating variable.

→ Example →

Amount of Insurance (AOI)	Territory	Exposures	Loss & ALAE	Premium at Current Rate Level
Low	1	10	$303.55	$567.60
Medium	1	110	$4,586.67	$7,804.50
High	1	180	$10,807.86	$17,368.56
Low	2	130	$6,416.59	$12,298.00
Medium	2	120	$8,136.00	$14,190.00
High	2	140	$13,668.48	$22,514.80
Low	3	150	$9,165.85	$18,447.00
Medium	3	120	$10,072.37	$18,447.00
High	3	40	$4,834.74	$8,362.64
TOTAL		1,000	$67,992.11	$120,000.10

The current base rate is $118.25. The base levels are Territory 2 and Medium AOI.

(long data)

→ First, organize the exposures + loss data as follows:

→ Exposures →

AOI \ Territory	1	2	3	Total
Low	10	130	150	290
Medium	110	120	120	350
High	180	140	40	360
Total	300	390	310	1,000

(wide data)

→ Loss + ALAE →

AOI \ Territory	1	2	3	Total
Low	$303.55	$6,416.59	$9,165.85	$15,885.99
Medium	$4,586.67	$8,136.00	$10,072.37	$22,795.04
High	$10,807.86	$13,668.48	$4,834.74	$29,311.08
Total	$15,698.08	$28,221.07	$24,072.96	$67,992.11

→ Equate the exposure-weighted loss costs, which is the observed loss + ALAE, + the indicated loss costs, which is the total premium calculated using the indicated relativities, for each level of each rating variable

$$\begin{aligned}
\text{Territory 1:}\quad & 15{,}698.08 = 118.25\,(10 \times a_L \times t_1 + 110 \times a_M \times t_1 + 180 \times a_H \times t_1)\\
\text{Territory 2:}\quad & 28{,}221.07 = 118.25\,(130 \times a_L \times t_2 + 120 \times a_M \times t_2 + 140 \times a_H \times t_2)\\
\text{Territory 3:}\quad & 24{,}072.96 = 118.25\,(150 \times a_L \times t_3 + 120 \times a_M \times t_3 + 40 \times a_H \times t_3)\\
\text{Low AOI:}\quad & 15{,}885.99 = 118.25\,(10 \times a_L \times t_1 + 130 \times a_L \times t_2 + 150 \times a_L \times t_3)\\
\text{Medium AOI:}\quad & 22{,}795.04 = 118.25\,(110 \times a_M \times t_1 + 120 \times a_M \times t_2 + 120 \times a_M \times t_3)\\
\text{High AOI:}\quad & 29{,}311.08 = 118.25\,(180 \times a_H \times t_1 + 140 \times a_H \times t_2 + 40 \times a_H \times t_3)
\end{aligned}$$

→ For one variable, say t_i, use seed values that are the univariate PP approach solutions
⇒ solve for all a_i using seed t_i values, then solve for t values using a values that we just found

→ Sequential analysis
→ Another technique related to the minimum bias procedure is sequential analysis. Sequential analysis is the only classification ratemaking method permitted for voluntary personal automobile insurance in California. It involves the following steps:
→ 1) Conduct a standard univariate analysis on the first variable to determine the indicated relativities
→ 2) Calculate the indicated relativities for the second variable using the adjusted pure premium approach. This involves adjusting the exposures based on the indicated relativities of the first variable obtained in step 1
→ 3) Repeat the adjusted pure premium approach for all remaining variables, each time adjusting the exposures based on the analysis of all prior variables.
→ 4) Conclude the process after completing a single pass through the sequence of chosen rating variables

→ Comparison
→ In contrast to minimum bias procedures, which iterate until convergence is achieved, sequential analysis is not iterative.
→ The main drawbacks of sequential analysis is that it does not have a closed-form solution, resulting in varying outcomes based on the order of rating variables in the sequence.

→ Sequential analysis → Pick first variable + find relativities to base like usual
⇒ Perform adj PP approach for next variable using results from above
⇒ repeat w/ new variable + new results
adjusts based on exposure distribution

⇒ If deciding which variables to adjust for in a univariate classification problem, think logically ⇒ don't have to "cross" every variable
⇒ use context + state decision

3.2.2 → Multivariate classification

→ Why needed?
→ Univariate methods
→ offer the advantage of transparency, but this benefit is outweighed by the inaccuracies of the method.
→ Results of univariate methods are not only distorted by distributional bias, but also by unsystematic effects (noise).
→ Commonly, univariate analysis lacks information about the certainty of results

→ Minimum bias procedures
→ Are an improvement over univariate approaches as they account for an uneven mix of business while maintaining transparency
→ Do not include diagnostics

→ Benefits of multivariate methods
→ 1) Take into account all rating variables simultaneously + automatically adjust for exposure correlations between rating variables
→ 2) Attempt to capture only systematic effects (signals) + ignore unsystematic effects (noise).
→ 3) Generate model diagnostics, offering additional information about the certainty of results + appropriateness of the fitted model
→ 4) Allow consideration of the interaction between two or more rating variables, known as response correlation

→ GLMs → Additional benefits
→ Transparent
→ model output includes parameter estimates for every level of each explanatory variable w/in the model, along w/ a range of statistical diagnostics

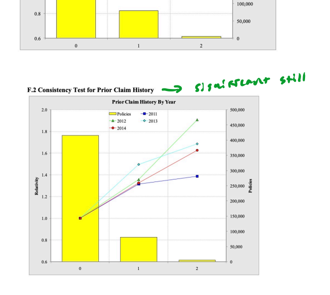
→ Data type used in analysis → typically performed on loss cost data, preferably frequency + severity data separately, instead of loss ratio data b/c:
→ 1) Modeling loss ratios requires on-level premiums at the granular level, which poses practical challenges.
→ 2) Experienced actuaries have an a priori expectation of frequency & severity patterns, making it easier to distinguish the signal from the noise, in contrast to LR patterns that are dependent on the current rates.
→ 3) Loss ratio models become obsolete when rates + rating structures are changed.
→ 4) There is no standard distribution for modeling loss ratios.

→ Uses of Diagnostics
→ 1) To determine if a predictive variable has a systematic effect on insurance losses + should thus be retained in the model.
→ 2) To evaluate the modeler's assumptions regarding the link function + the error term.

→ Common diagnostics
→ 1) Standard errors → measure variability associated w/ a parameter + often used to calculate the CI of a point estimate
→ 2) Measures of deviance → Deviance is a summary measure of the extent to which the fitted values deviate from the actual observations. Examples of deviance tests are chi-square, F-test, AIC, BIC
→ 3) Practical diagnostics → An example is to run the GLM for individual years of a multi-year dataset to assess the consistency of the results over time

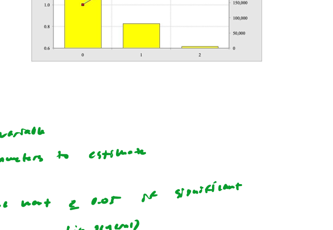
→ Model validation
→ Other than running statistical diagnostics, an analyst should also evaluate how well the model validates. One example of validation techniques is comparing the model's estimation w/ historical results on a holdout sample of data, which are not used in the development of the model. This can identify if the model is over- or under-fitting.

→ Actual test to run through when determining the predictive power of a potential rating variable w/ DECISION:



→ Main effects test (parameters + st errors):
→ looking for up/downward sloping indicated relativity lines w/ tight st errors (keeping in mind volume of data)

→ Consistency test:



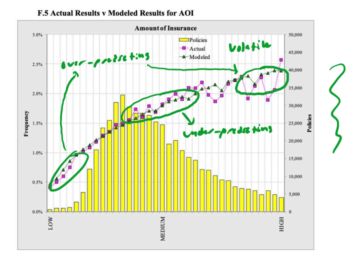
→ checks the result of the GLM to see if all years have a constant slope + are consistent over the years (i.e. want consistent predictions over time)
→ Looking for consistent (non-flat) slopes over years
⇒ If so, then have evidence supporting the variable's predictive power

→ Statistical test
→ Just testing significance through a regression model
⇒ So, fitting model w/ + w/o variable of interest + analyzing the trade-off between increased accuracy of the model w/ the variable included vs the additional complexity of having additional parameters to estimate
→ e.g chi-squared test, T-test, F-test
↳ will be given "a chi-square percentage", which we want ≤ 0.05 if significant (in general)

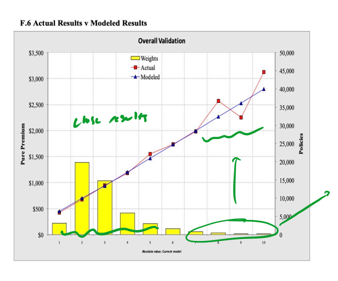
→ Judgemental
→ check to see if relativities make common sense + an intuition
→ e.g) frequency is higher w/ increased claims

→ Final decision → All four tests suggest the rating variable is predictive + should be included in the model (+ ultimately the rating algorithm)

→ Overall model validation → comparing predictions made by model to actual results on a holdout dataset: validation tests
→ (...) Segmented by variable
→ If the model is predictive, then the response should be very close for any level w/ enough volume to produce stable results
→ The random nature of the insurance process will create small differences between the lines; however, either large or systematic differences or both should be investigated as possible indicators of an ineffective model.
⇒ e.g) the model may contain too much noise caused by retaining statistically insignificant variables or not have enough explanatory power b/c statistically significant variables are omitted.



→ (...) Segmented by fitted value
→ underlying freq + sev models were used to determine the modeled pure premium for each observation in the holdout dataset
→ Then, each obs was ordered according to modeled PP from lowest to highest expected value
→ Then grouped
→ Then compared modeled + actual results



→ GLMs correct for (take into account) correlation between variables + exposure distribution bias, + interaction